

Building a Retirement Income TIPS Ladder

A TIPS ladder can be constructed similarly to a Treasury STRIPS ladder, also using data from the *Wall Street Journal* Market Data Center, which provides a daily report of wholesale prices from the secondary markets for all outstanding TIPS issues. On January 3, 2017, there were forty outstanding TIPS available, with maturity dates ranging from January 2017 to February 2046. As mentioned before, there are no TIPS maturing in 2030–31 and 2033–39. For these years, we buy extra TIPS for the most recent maturing year (i.e., 2029 for 2030–31 and 2032 for 2033–39) to cover the later spending, assuming those proceeds subsequently grow at a 0 percent real return until they are needed for spending.

TIPS provide coupon payments, which make the calculations a bit more complex than that of STRIPS. To construct a ladder with coupon bonds, we have to work backward from the ladder's end date. Our goal is to construct a TIPS ladder that provides \$40,000 of inflation-adjusted income for thirty years between 2017 and 2046. Exhibit 7.12 shows the process for constructing this TIPS ladder.

Exhibit 7.12

Constructing a 30-Year TIPS Ladder on January 3, 2017, for an Annual Real Income of \$40,000

Maturity Date	Coupon Rate	Ask Price	Accrued Principal	Nominal Ask Price	Real Yield to Maturity (%)	Real Targeted Spending Amount	# of Shares Purchased	Cost of Shares
2017	0.00%	\$100.00	\$1,000	\$1,000	0	\$40,000	22.27	\$22,266
2018	1.63%	\$102.63	\$1,153	\$1,183	−0.91%	\$40,000	19.31	\$22,851
2019	2.13%	\$105.63	\$1,125	\$1,188	−0.62%	\$40,000	20.11	\$23,901
2020	1.38%	\$105.25	\$1,117	\$1,176	−0.34%	\$40,000	20.69	\$24,322
2021	1.13%	\$105.09	\$1,104	\$1,160	−0.14%	\$40,000	21.22	\$24,620
2022	0.13%	\$100.38	\$1,067	\$1,071	0.05%	\$40,000	22.2	\$23,779
2023	0.13%	\$99.50	\$1,047	\$1,042	0.21%	\$40,000	22.66	\$23,601
2024	0.63%	\$102.06	\$1,035	\$1,056	0.33%	\$40,000	22.95	\$24,240